Predicting 30-Day Hospital Readmission Among Patients with Diabetes

Siddique Chand, Aubin Mugisha, and Cameron Letendre
University of Southern Maine, Portland, ME, USA

***Abstract*—We studied which machine learning models can predict 30-day hospital readmission among patients with diabetes. After cleaning the Diabetes 130-US Hospitals dataset, we compared logistic regression, random forest, and XGBoost on 99,340 stays. We split the data by patient so that all stays from the same patient were in one set. Random forest had the highest ROC-AUC (0.669), and XGBoost had the highest average precision (0.221) and recall (0.581). Both improved ROC-AUC only slightly over logistic regression. Previous inpatient visits and discharge destination were the most useful predictors for both tree models. At a cutoff of 0.5, the models found about half of the recorded early readmissions. Fewer than one in five flagged stays was followed by readmission.**

I. Introduction

A 30-day readmission happens when a patient returns to the hospital for another stay within 30 days of discharge. Some returns are planned. Others happen when a health problem continues or a new one develops after discharge. Readmissions also add to hospital costs. In 2012, Medicare began reducing payments to hospitals with excess readmissions under the Hospital Readmissions Reduction Program [1].

Patients with diabetes are readmitted more often than the general hospital population. Rubin [2] reports 30-day readmission rates of 14.4% to 22.7% for patients with diabetes, compared with 8.5% to 13.5% for all hospitalized patients. Many patients with diabetes also have conditions such as heart or kidney disease and take several medications. This can make their care after discharge more difficult to manage.

If a care team could identify patients who are more likely to return, it could use that information to plan a follow-up call, an earlier clinic visit, or a medication review. In this project, we use the Diabetes 130-US Hospitals dataset [3] to predict whether a hospital stay will be followed by readmission within 30 days. We ask two questions:

1) How well do logistic regression, random forest, and XGBoost predict 30-day readmission on this dataset?
2) Which patient and hospital-stay factors are most useful for predicting readmission?

Only about 11% of stays were followed by readmission within 30 days. In our evaluation, we look at how many of these readmissions each model finds and how often its predictions of readmission are correct.

II. Literature Review

A. Clinical Factors Related to Readmission

Soh et al. [4] reviewed 23 studies on unplanned readmission among patients with diabetes. They found that heart failure, kidney disease, insulin treatment, and insurance status were linked to readmission within 30 days. These findings help explain why our study includes information about other diagnoses, medications, and insurance.

Strack et al. [5] studied 69,984 hospital stays, using one stay per patient. They found that the relationship between HbA1c testing and early readmission varied by the patient's main diagnosis. They also considered changes in diabetes medication. In our project, we record whether an HbA1c test was done and whether testing and a medication change occurred during the same stay.

B. Comparing Prediction Models

Shang et al. [6] compared random forest, naive Bayes, and a tree ensemble on the same hospital dataset we use. Random forest performed best among the three. Previous hospital admissions, emergency visits, age, and diagnoses were among the useful predictors. Their findings give us a reason to look closely at how often patients used the hospital before their current stay.

Liu et al. [7] compared eleven models on the same dataset. They reported ROC-AUC scores of 0.64 for XGBoost, 0.63 for random forest, and 0.61 for an LSTM neural network. The small differences between these scores give us a reference for comparing models on these data.

Hai et al. [8] used more detailed records from patients' previous stays to predict unplanned readmission. Their LSTM reached a ROC-AUC of 0.79, compared with 0.72 for random forest. They also reported better LSTM performance when more patient history was available. Their data and outcome differ from ours, so the scores are difficult to compare directly.

C. Evaluating Readmission Predictions

Emi-Johnson and Nkrumah [9] reported about 89% accuracy for XGBoost. Their table for the readmitted class reported a recall of 0.017, meaning that fewer than two out of every 100 actual readmissions were identified. Some figures elsewhere in their paper differ from those in the table, so we treat this as an example of what overall accuracy can hide.

Saito and Rehmsmeier [10] explain why it helps to look at precision and recall together when one outcome is much less common than the other. In our data, about 11% of stays were followed by readmission within 30 days. We therefore look at how many of those readmissions the models find and how often their flags are correct, alongside how well they rank stays by readmission risk.

III. Dataset

We use the Diabetes 130-US Hospitals for Years 1999–2008 dataset from the UCI Machine Learning Repository [3]. It contains 101,766 hospital stays from 71,518 patients at 130 US hospitals. Each row is one stay that lasted 1 to 14 days and included a diabetes diagnosis, laboratory tests, and medications [5]. Some patients had several stays, so they appear in more than one row.

The dataset has 50 columns: 8 numeric, 40 categorical (including the outcome), and 2 IDs. They describe the patient (age, race, gender), the stay (admission type, length of stay, diagnoses, discharge destination), lab results, diabetes medications and dose changes, and the number of outpatient, emergency, and inpatient visits in the year before the stay.

The outcome column, `readmitted`, shows whether the patient was readmitted within 30 days (`<30`), after more than 30 days (`>30`), or not at all (`NO`). Since we focus on early readmission, we label `<30` as 1 and the other two values as 0.

IV. Exploratory Data Analysis

A. Descriptive Statistics

Table I summarizes the numerical variables before cleaning. Stays averaged 4.4 days, and 64.6% involved patients aged 60 to 89.

TABLE I
DESCRIPTIVE STATISTICS FOR NUMERICAL VARIABLES

Variable	Mean	SD	Median	Range
Days in hospital	4.40	2.99	4	1–14
Laboratory procedures	43.10	19.67	44	1–132
Other procedures	1.34	1.71	1	0–6
Medications	16.02	8.13	15	1–81
Previous outpatient visits	0.37	1.27	0	0–42
Previous emergency visits	0.20	0.93	0	0–76
Previous inpatient visits	0.64	1.26	0	0–21
Diagnoses	7.42	1.93	8	1–16

SD = standard deviation.

The previous-visit columns describe the year before each hospital stay. For outpatient visits, 83.6% of records showed zero visits. The share with zero visits was 88.8% for emergency visits and 66.5% for inpatient visits. This explains why the median is zero for all three columns. Some records still showed frequent hospital use. The highest recorded emergency visit count was 76.

Some patients had more than one hospital stay in the dataset, so they appeared in more than one row. This applied to 16,773 of the 71,518 patients (23.5%). Together, their stays accounted for 46.2% of all records. We used the patient ID to keep each patient's stays together when splitting the data. This lets us test the models on patients outside the training set.

B. Readmission Patterns

About one in nine stays was followed by readmission within 30 days (Fig. 1(a)). A model that always predicted no early readmission would be right 88.8% of the time and still miss every early readmission.

TABLE II
30-DAY READMISSION RATE BY DISCHARGE DESTINATION

Discharge group	Stays	Rate (%)
Home	60,232	9.3
Home with care	13,010	12.7
Nursing facility	14,817	14.6
Hospital	2,160	16.2
Rehab/LTC/Psych	3,790	24.0
Left against medical advice	623	14.4
Unknown	4,708	11.7

Excludes stays ending in death or hospice.

On the other hand, readmission was more common among patients who had been admitted more often in the previous year. The rate rose from 8.4% for stays with no previous inpatient visits to 20.3% for stays with three (Fig. 1(b)). The groups with the most previous visits contained few stays, so we interpret their rates with more caution. Previous emergency visits showed a similar pattern.

Readmission was also more common after longer stays (Fig. 1(c)). For insulin, rates were higher when the dose went up or down than when it stayed steady (Fig. 1(d)).

Readmission rates also varied by where patients went after discharge (Table II). The rate was lowest for patients sent home and highest for those sent to rehabilitation, long-term care, or psychiatric facilities.

C. Missing Values

Weight was missing from nearly every stay. Medical specialty and payer code were also often missing. Figure 2 shows how common missing values were in each column.

Lab results were recorded for relatively few stays. About one in six stays had an HbA1c result, and about one in twenty had a serum glucose result. During cleaning, we kept track of whether each result was recorded.

V. Data Cleaning and Preparation

We prepared the data in Python using pandas.

A. Removed Records and Columns

Following Strack et al. [5], we removed 1,652 stays that ended in death and 771 that ended in discharge to hospice. We also removed three stays where gender was recorded as "Unknown/Invalid." This left 99,340 stays from 69,987 patients.

We dropped weight, which was missing in almost 97% of rows. We also dropped nine drug columns that were almost never used like examide and citoglipton which were never given, and the other seven were given in just 23 stays or fewer. We kept the stay and patient IDs in a separate file. They are not used as predictors, but the patient ID lets us keep each patient's stays together when we split the data.

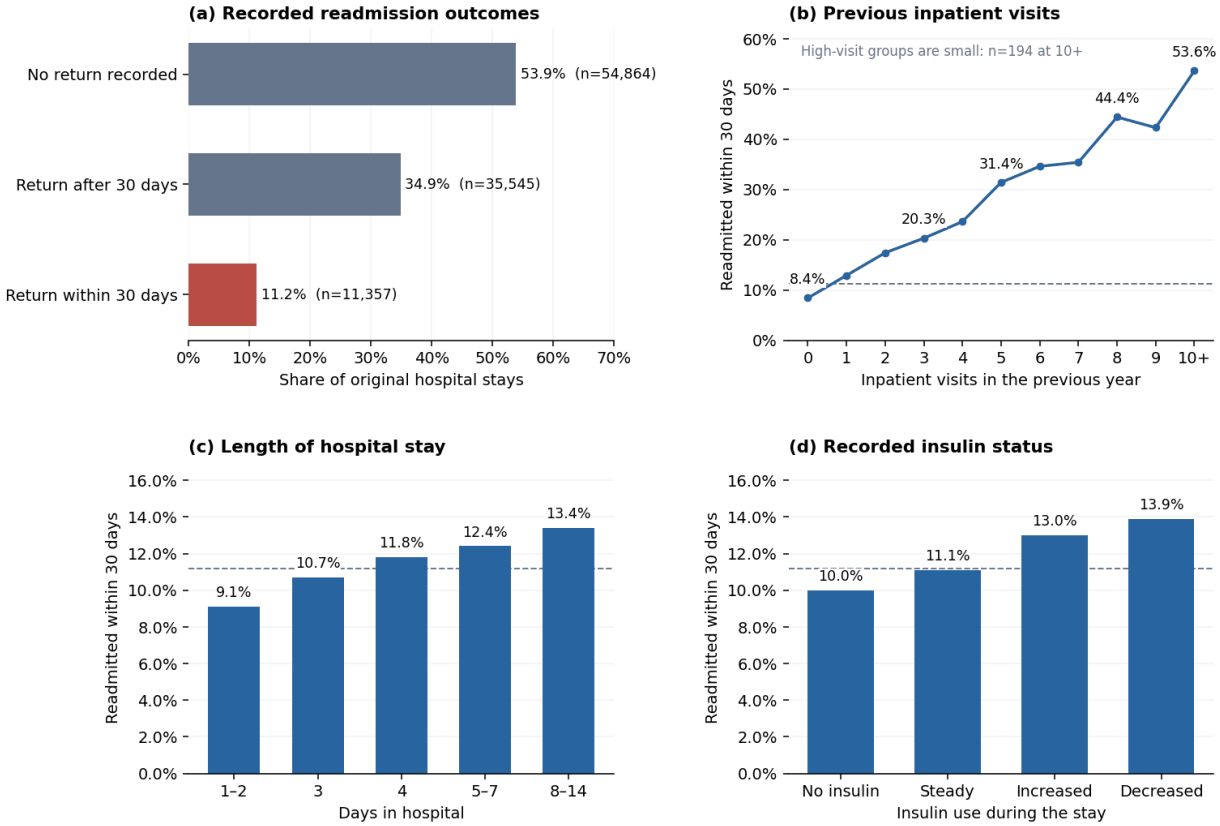


Fig. 1. Readmission patterns in the original dataset. Panel (a) shows the recorded outcomes. Panels (b)–(d) show 30-day readmission rates by previous inpatient visits, length of stay, and insulin status. The dashed line marks the overall rate of 11.2%.

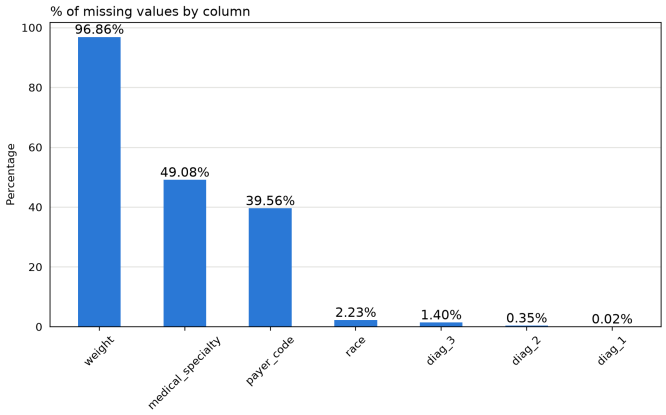


Fig. 2. Percentage of stays with missing values in each of the seven columns shown.

B. Missing Values

We used a "Missing" category when race, payer code, medical specialty, or a diagnosis code was missing. When there was no HbA1c or glucose result, we used "Not measured."

We grouped admission and discharge codes such as "NULL" and "Not Mapped" under "Unknown." We also put newborn and birth-related codes in that group because 13 of the 14 stays with those codes were for adults. All numerical columns were complete.

C. Coding and Grouping Categories

Some categorical columns had too many values to use directly. Many values appeared in only a few stays, which gives a model too little to learn from. We combined these values into larger groups with a clear meaning.

1) Admission and Discharge Codes: These columns use numbers to describe how a patient entered or left the hospital. We used the dataset's mapping file to identify what each code meant and combine similar ones. For example, codes for different types of transfers were combined into one transfer group. Table III shows the resulting groups.

2) Diagnosis Codes: Each of the three diagnosis columns contained more than 700 ICD-9 codes. We combined them into nine broader diagnosis groups using the approach described by Strack et al. [5]. For example, codes from 390 to 459 were placed in the circulatory group. Codes beginning with V or E were placed in "Other," and missing diagnoses remained in a separate category.

We applied the same rules to all three diagnosis columns. Circulatory conditions were the most common primary diagnosis group, accounting for about 30% of stays.

3) Medical Specialty: This column identifies the admitting doctor's specialty. It contained 73 categories, including

TABLE III
GROUPS FOR ADMISSION AND DISCHARGE CODES

Column	Codes	Groups
Admission type	8	Emergency, urgent, elective, unknown
Admission source	17	Emergency room, referral, transfer, unknown
Discharge	20*	Home, home with care, nursing facility, rehab/long-term/psychiatric care, other hospital, left against medical advice, unknown

*After removing death and hospice codes.

"Missing." Of these, 44 appeared in fewer than 100 stays. We kept the ten most common categories, which covered 91.8% of stays, and combined the remaining categories into "Other."

4) Age: Age was recorded in ten-year groups. We numbered these groups from 0 to 9: ages 0–9 became 0, ages 10–19 became 1, and so on. This kept the age groups in order using a single column.

VI. Methods

A. Models and Inputs

We used logistic regression as our baseline. Of the 14 drug columns, we kept insulin because it was the most common drug and readmission rates varied with its dose status. This left 29 predictors. We scaled the numerical predictors to a mean of 0 and a standard deviation of 1. For categorical predictors, we created a separate 0-or-1 input for each category using one-hot encoding. This gave 111 model inputs.

Random forest averages predictions from many decision trees. XGBoost builds trees one at a time to improve earlier predictions. Both models used all 42 predictors, including the 14 drug columns. We kept the original numerical values and used the same type of encoding for categories, giving 161 inputs.

We gave early readmissions more weight during training because they were much less common. Logistic regression and random forest used balanced class weights. For XGBoost, we set the weight for early readmission to 7.78, the ratio of stays without early readmission to stays with early readmission in the training set.

B. Training and Data Split

We used scikit-learn's `StratifiedGroupKFold` to create five folds with a random seed of 42. This kept each patient's stays together and gave the folds similar readmission rates. We used the first fold for testing and the other four for training. The training set had 79,473 stays from 56,036 patients, and the test set had 19,867 stays from 13,951 patients. Both sets had a readmission rate of about 11.4%.

All three models used the same split. We used fixed settings for logistic regression and tested four combinations of settings for each tree model. We compared those combinations using three-fold cross-validation within the training set, grouping stays by patient. We chose the settings with the highest average ROC-AUC and used the test set for the final comparison (Table IV).

TABLE IV
MODEL SETTINGS

Model	Settings
Logistic regression	L2 penalty, $C = 1.0$, `lbfgs` solver, balanced class weights, maximum 2,000 iterations
Random forest	300 trees, maximum depth 20 (tested 10 and 20), minimum leaf size 10 (tested 10 and 50), balanced class weights
XGBoost	300 trees, maximum depth 3 (tested 3 and 5), learning rate 0.05 (tested 0.05 and 0.1), subsample 0.8, positive-class weight 7.78

C. Evaluation

ROC-AUC measures how well a model ranks stays followed by readmission above other stays. Average precision (AP) summarizes precision and recall across cutoffs. For random predictions, the reference scores are 0.5 for ROC-AUC and about 0.114 for AP, the test-set readmission rate.

At a cutoff of 0.5, recall tells us how many early readmissions a model found, and precision tells us how often its flags were correct. F1 combines these two measures. We also compared the models when each flagged the 20% of stays with the highest scores. This gave each model the same follow-up workload.

For the tree models, we measured how much AP dropped when we shuffled one predictor at a time in the test set. We also compared ROC-AUC across insurance groups and repeated the logistic regression evaluation on five patient splits to see how much its score varied.

VII. Results

A. Model Performance

TABLE V
MODEL PERFORMANCE ON THE TEST SET

Model	Precision	Recall	F1	ROC-AUC	AP
Logistic regression	0.179	0.542	0.270	0.658	0.214
Random forest	0.198	0.492	0.282	0.669	0.214
XGBoost	0.179	0.581	0.273	0.666	0.221

Precision, recall, and F1 use a cutoff of 0.5.

The ROC-AUC scores were close, ranging from 0.658 to 0.669. Random forest had the highest ROC-AUC, and XGBoost had the highest AP. Figure 3 shows how the three models compared across cutoffs.

At the 0.5 cutoff, logistic regression flagged 6,836 of the 19,867 test stays and found 1,227 of the 2,263 early readmissions. It found a little over half of the readmissions, and fewer than one in five of its flags was correct.

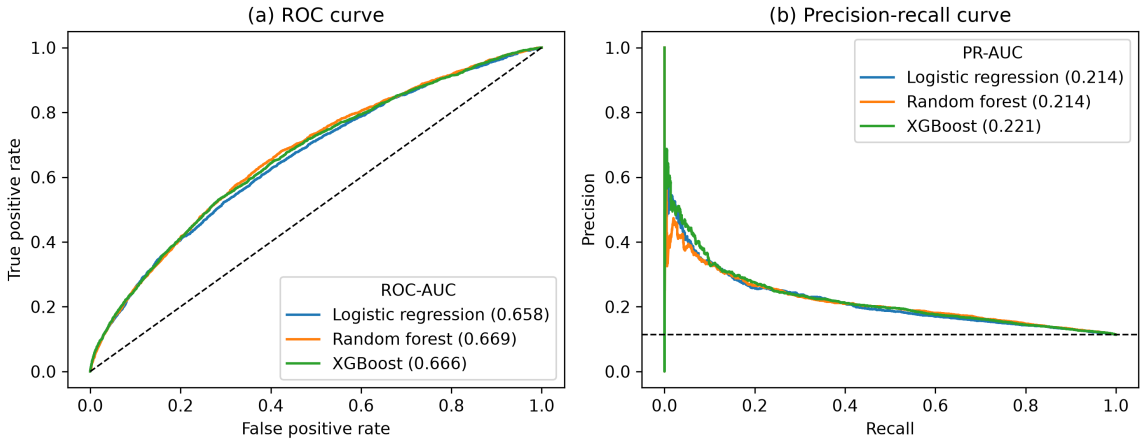


Fig. 3. Test-set curves for the three models. Panel (a) shows the ROC curves. Its dashed line marks random ranking. Panel (b) shows the precision–recall curves. Its dashed line marks the test-set readmission rate (0.114). The legends report ROC-AUC and AP.

Random forest had the highest precision. It flagged 28.3% of the test stays and found 1,113 early readmissions. XGBoost flagged 37.0% and found 1,314, giving it the highest recall at this cutoff. It also produced more false alarms.

Across five patient splits, logistic regression had a mean ROC-AUC of 0.660, with scores from 0.653 to 0.666. Changing the split moved its score about as much as changing the model on the main test set.

B. Comparison at Equal Workload

When each model flagged the highest-scoring 20% of stays, the results were similar. Precision ranged from 0.213 to 0.217, and recall ranged from 0.374 to 0.381. About one in five flagged stays was followed by early readmission, compared with about one in nine under random flagging. At this workload, the models found nearly the same share of early readmissions.

C. Important Predictors

Previous inpatient visits and discharge destination had the largest drops in AP when shuffled in both tree models (Table VI). Shuffling either predictor reduced AP by at least 0.030. For every other predictor, the drop was at most 0.006. Both variables also showed clear differences in readmission rates in Section IV-B.

In logistic regression, a one-standard-deviation increase in previous inpatient visits multiplied the fitted odds of readmission by 1.44, with the other inputs held fixed. We used both discharge coefficients to compare groups. With the other inputs held fixed, the fitted odds for discharge to rehabilitation, long-term care, or psychiatric facilities were about 3.3 times those for discharge home ($2.10/0.64$).

D. Performance by Insurance Group

All three models had their lowest ROC-AUC for Medicare stays, with scores from 0.642 to 0.656. Medicare stays also had a higher readmission rate than stays with other or missing insurance information (12.4% versus 10.8%). The Medicaid group had only 698 test stays, so we interpret its scores with more caution.

TABLE VI
DROP IN TEST AP AFTER SHUFFLING EACH PREDICTOR

Predictor	Random forest	XGBoost
Previous inpatient visits	0.046	0.068
Discharge destination	0.036	0.030
Primary diagnosis group	0.003	0.005
Medical specialty	0.005	0.003
Secondary diagnosis group	0.001	0.003
Number of diagnoses	0.002	0.001

E. Class Imbalance and Decision Threshold

Grouped cross-validation on the training set gave these results:

- Class weights: AP 0.217, ROC-AUC 0.673
- Random oversampling: AP 0.216, ROC-AUC 0.671
- SMOTENC: AP 0.138, ROC-AUC 0.572

Class weights and random oversampling performed almost the same, and SMOTENC performed much worse, so we kept class weights.

We then chose the cutoff with the highest F1 on out-of-fold predictions from the training set. This cutoff was 0.513, close to the default of 0.5. On the test set:

- At 0.5: precision 0.198, recall 0.492, F1 0.282; 28.3% of stays flagged
- At 0.513: precision 0.203, recall 0.439, F1 0.278; 24.6% of stays flagged
- At 0.513, the model found 993 of the 2,263 early readmissions

Balanced class weights had already placed the best balance of precision and recall near 0.5, so tuning the cutoff mainly traded recall for fewer flagged stays.

TABLE VII
ROC-AUC ON THE DIABETES 130-US HOSPITALS DATASET

Model	This study	Shang et al. [6]	Liu et al. [7]
Logistic regression	0.658	–	0.63
Random forest	0.669	0.661	0.63
XGBoost	0.666	–	0.64
LSTM	–	–	0.61

A dash indicates no reported score for that model.

VIII. Discussion

A. Comparison With Earlier Studies

Table VII compares ROC-AUC scores for models trained on the same source dataset. Our random forest score was close to Shang et al.'s, and our XGBoost score was slightly higher than Liu et al.'s. The studies prepared and tested the data differently, so the scores give us a rough comparison.

Hai et al. [8] used a different dataset with more patient history and focused on unplanned readmissions.

B. What the Results Mean

Strack et al. [5] studied the link between HbA1c testing and readmission using one stay per patient. We made a similar sample by keeping the stay with the lowest encounter ID for each patient. This gave 69,987 stays and a 9.0% readmission rate, close to their 69,984 stays and 9.2% rate. The additional stays from patients with repeat admissions had a readmission rate of 17.1%. Including these stays raised the overall rate in our sample to 11.4%.

As in Strack et al.'s study, readmission was less common when an HbA1c test was recorded (9.9% versus 11.7%). The HbA1c indicators made a small contribution to our predictions. Previous inpatient visits and discharge destination were more useful to the models. Both are available by discharge and may reflect differences in how much care patients needed.

The tree models gave small gains in ROC-AUC, and all three models found a similar share of readmissions when they flagged the same number of stays. If staff have time for only a limited number of follow-up calls, the equal-workload comparison shows what each model would find at that limit.

C. Limitations

The data cover 1999 to 2008 and use ICD-9 codes. Changes in hospital care and coding may affect how well the models work on newer data. The recorded outcome combines planned and unplanned readmissions. Some early readmissions may be missing because they happened at hospitals outside the database.

We tested the tree models on one patient split and compared four settings combinations per model. The models' ranking and classification scores tell us how well they identified readmissions in this test set. A separate calibration check would show how closely their predicted probabilities match observed readmission rates.

IX. Future Work

We could repeat the tree model comparisons across more patient splits to see whether the close results hold. Newer hospital records with more detail about previous stays would let us test whether fuller patient histories improve prediction, as suggested by the results of Hai et al. [8]. We could also check the predicted probabilities against observed rates and compare the models at different follow-up workloads.

X. Conclusion

We compared three models for predicting 30-day hospital readmission among patients with diabetes. Random forest did slightly better than logistic regression at ranking stays by risk. XGBoost found the most readmissions in our first comparison, but it also gave us the longest follow-up list. When we limited each list to 20% of the test stays, all three models found a little over a third of the readmissions.

We also found that readmission was more common among patients who had already spent time in the hospital. The rate was about 8% for stays with no hospital admissions in the previous year and about 20% for stays with three. Previous hospital stays and where patients went after discharge were the two most useful pieces of information for both tree models. A next step would be to see whether more detailed records from those earlier stays help the models find more readmissions with the same number of follow-up calls.

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